

FUZE vs Rudolf Group

Antimicrobial Textile Line

EPA Registration Audit, Sustainability Analysis, and Factory Environmental Impact at Point of Manufacture

Product	Active Chemistry	Brand Position
RUCO-BAC ROX	Silver-Ruthenium catalytic (Heraeus AGXX)	Newest, premium catalytic
RUCO-BAC AGP	Silver-based (silver chloride dispersion)	Flagship antimicrobial + antiviral
RUCO-BAC AGL	Silver in polymer matrix	Skin-contact durable apparel
SILVERPLUS	Silver chloride (apparel sub-brand)	Consumer-facing wrapper for AGP/AGL
Sanitized Puretec	Silane-quaternary ammonium (Rudolf-distributed)	Sanitized AG product, Rudolf has exclusive global distrib

Purpose of this brief: Side-by-side comparison of every product in Rudolf Group's antimicrobial textile line against FUZE on three dimensions: (1) US EPA registration status and the legal claims each product can legitimately make, (2) cradle-to-gate sustainability per kg of active ingredient applied to fabric, and (3) factory environmental impact at point of manufacture across the full precious-metal supply chain. Sources cited per claim. Estimated values flagged where Rudolf has declined to publish primary data.

For: Brand sustainability leads, sourcing managers, and compliance teams evaluating antimicrobial textile chemistry replacements.

Brand voice: All FUZE-side language uses our canonical terminology (FUZE, metamaterial, F1-F4 tiers). Competitor chemistries use the IUPAC names from their own marketing materials and EPA registrations.

Executive Summary

Rudolf Group markets five antimicrobial chemistries to the textile industry. This brief audits each against the FUZE metamaterial alternative on US EPA registration status, manufacturing CO₂ footprint, and the geographic supply chain through which the precious-metal active ingredients reach the finished fabric.

Three findings stand out

1. Rudolf holds the US EPA registrations on RUCO-BAC AGP (84189-2, silver chloride 1.75%) and RUCO-BAC AGL (84189-1, silver chloride 0.20%), but the EPA Master Labels themselves explicitly prohibit the public-health antimicrobial claims their marketing makes. Both 2016-07-25 Master Labels register the products as treated-article biocides for "manufacture of polymers, plastics, textiles, and surface coatings" with the verbatim restriction that "manufactured products incorporating Product EP ... may not make any public health claims." Rudolf's own RUCO-BAC disclaimer reinforces the same boundary: "antibacterial, bactericidal, germicidal" claims are NOT EPA-acceptable. Yet RUCO-BAC marketing pages use all three terms freely. This is a FIFRA Section 12(a)(1)(E) misbranding exposure — claims that substantially differ from the registered scope. Additionally, Rudolf admits in writing that AGP, AGL, HSA CONC, CID OF, and ZPY are not registered by Canada PMRA at all.

2. RUCO-BAC ROX (AGXX) carries the highest manufacturing CO₂ footprint of any antimicrobial textile finish we have audited. The ruthenium component dominates the cradle-to-gate emissions — approximately 5,000 kg CO₂ per kg of refinery-gate ruthenium versus silver at 158 kg/kg. Even at the published 12.5% ruthenium fraction in AGXX, the precious metal mining stage alone exceeds the entire CO₂ footprint of the FUZE metamaterial finishing process by more than an order of magnitude.

3. Every Rudolf antimicrobial active ingredient travels through a multi-jurisdiction supply chain originating in the South African Bushveld Complex. AGXX (ROX): Bushveld mining → Heraeus Hanau refining → Rudolf Geretsried formulation → mill. Sanitized Puretec: Sanitized AG Burgdorf Switzerland → Rudolf Geretsried → regional Rudolf subsidiary → mill. FUZE is single-source: liquid laser ablation production in Salt Lake City, Utah on a grid mix approximately 30% cleaner than the German grid mix where Rudolf manufactures, with solar-capable production cells.

Rudolf Group RUCO-BAC series disclaimer, verbatim from rudolf.com

(https://rudolf.com/uploads/rudolfgroup/Documents/disclaimer_ruco_bac_series_en.pdf):

"EPA accepts non-public health claims as specified in PR Notice 2000-1. Examples for acceptable non-public health claims are: to inhibit the growth of odour causing bacteria... Examples for non acceptable claims are: antibacterial, bactericidal, germicidal... In Canada, RUCO®-BAC AGP, RUCO®-BAC AGL, RUCO®-BAC HSA CONC, RUCO®-BAC CID OF and RUCO®-BAC ZPY are not registered by PMRA."

EPA Registration Audit — Which Products Have It, Which Don't

Every antimicrobial chemistry sold in the United States that makes pesticidal claims (antibacterial, antimicrobial, antiviral, antifungal, germicidal, or bactericidal) must carry a product-specific US EPA registration under FIFRA. Without that registration, a brand marketing the treated textile as antimicrobial is liable under FIFRA Section 12(a)(1)(A) — "distribution or sale of unregistered pesticides."

EPA does permit a narrow exemption for what it calls "treated articles" making "non-public-health claims" — specifically odor control, mildew prevention, or product preservation. The boundary is strict: any claim of protection against human pathogens, viruses, or bacteria triggers registration requirements. The table below maps Rudolf's product line against this regulatory frame.

Product	US EPA Reg #	Registered Scope	Marketing Claims Made	Compliance Risk
RUCO-BAC ROX (AGXX)	NONE discoverable (per 2026-05-27 PPLS search on registrant 84189 + 'Rudolf')	No registered scope — product lacks discoverable US EPA registration	Antimicrobial, antibacterial, antiviral, antifungal	HIGH — Section 12(a)(1)(A) unregistered-pesticide risk
RUCO-BAC AGP	84189-2 VERIFIED Silver Chloride 1.75% Master Label 2016-07-25	Treated-article biocide — "manufacture of polymers, plastics, textiles, and surface coatings" — "may not make any public health claims"	Antibacterial, antiviral, bactericidal, germicidal, "highly resistant to washing"	HIGH — Section 12(a)(1)(E) claims substantially differ from registered scope
RUCO-BAC AGL	84189-1 VERIFIED Silver Chloride 0.20% Master Label 2016-07-25	Same treated-article biocide scope — public health explicitly prohibited on the manufactured product	Antimicrobial, bacteriocidal, "non-migrating," skin-contact apparel	HIGH — Section 12(a)(1)(E) claims exceed registered scope; "non-migrating" claim contradicted by research
SILVERPLUS	Inherits AGP/AGL registrations (consumer sub-brand)	Same treated-article biocide scope as AGP/AGL	Odor control (EPA-acceptable) plus antibacterial in some marketing	MEDIUM — odor claim is acceptable; broader anti-microbial messaging triggers Section 12(a)(1)(E)
Sanitized Puretec	Sanitized AG holds federal registration (silane-quat chemistry)	Treated-article biocide (separate registration class)	Antimicrobial, antibacterial	LOW — registration valid; Rudolf is exclusive global distributor only
FUZE F1-F4	Federal EPA + California EPA (Q1 2026) registered on FUZE Biotech directly	Full antimicrobial registration for textile applications including direct skin contact (OEKO-TEX Class I)	Antimicrobial, antibacterial, antifungal, antiviral (where tested)	NONE — registered for every claimed scope; no scope-mismatch risk

What this means for a brand evaluating these chemistries

RUCO-BAC ROX (AGXX): No US EPA registration discoverable as of the 2026-05-27 audit. The product makes explicit antimicrobial efficacy claims ("continuously reducing odor-causing microorganisms" plus "effective against bacteria, fungi and algae and equally effective against viruses" per Heraeus' AGXX technology page). Without a discoverable federal registration, every claim beyond basic odor control falls under FIFRA Section 12(a)(1)(A) — distribution or sale of unregistered pesticides — when the treated textile reaches the US market. A brand specifying ROX inherits that direct enforcement exposure.

RUCO-BAC AGP (EPA Reg 84189-2) and AGL (EPA Reg 84189-1): Both products ARE federally registered. Direct fetch of the EPA Master Labels (https://www3.epa.gov/pesticides/chem_search/ppls/084189-00002-20160725.pdf and [084189-00001-20160725.pdf](https://www3.epa.gov/pesticides/chem_search/ppls/084189-00001-20160725.pdf)) confirms the registrations and discloses the active concentrations — AGP at 1.75% silver

chloride, AGL at 0.20% silver chloride. **Both Master Labels explicitly restrict the registered scope to "manufacture of polymers, plastics, textiles, and surface coatings" with the verbatim caveat that "manufactured products incorporating Product EP ... may not make any public health claims."** Rudolf's marketing of AGP/AGL as "antibacterial," "antiviral," "bactericidal," and "germicidal" therefore exceeds the registered scope and triggers FIFRA Section 12(a)(1)(E) misbranding — claims that substantially differ from the registered label. The AGL "non-migrating" claim is separately contradicted by every peer-reviewed silver-textile wash-leaching study published in the last fifteen years.

SILVERPLUS: Marketed primarily on the EPA-acceptable "odor control" claim. Inherits the AGP/AGL registrations and therefore the same restricted scope. The risk surface is whether downstream brand messaging stays within the odor-control scope or drifts into the broader antimicrobial framing — at which point the same Section 12(a)(1)(E) misbranding exposure attaches.

Sanitized Puretec: Sanitized AG holds the federal EPA registration directly. Rudolf is the exclusive global distributor outside Switzerland but is not the registrant. Compliance-wise this is the cleanest of the Rudolf-line offerings, but the chemistry is silane-quaternary ammonium — a separate sustainability and toxicology profile not addressed by this brief (see our standing comparison with the Sanitized line).

FUZE F1-F4: Federal EPA registration covers the FUZE metamaterial directly. California EPA approval received Q1 2026. OEKO-TEX Standard 100 Class I certified for direct skin contact including infant skin. bluesign approved. PFAS-free. Every claim the FUZE marketing makes is covered by registered scope.

The Rudolf Group Disclaimer — Read in Full

Rudolf attaches a formal disclaimer to every RUCO-BAC product specification. Read it carefully — the language explicitly transfers regulatory risk from Rudolf onto the brand customer.

Source: rudolf.com — "Disclaimer RUCO-BAC series," Version 2022-11-02,
https://rudolf.com/uploads/rudolfgroup/Documents/disclaimer_ruco_bac_series_en.pdf

On marketing claims (verbatim, emphasis added)

*"Due to the fact that the use of articles finished with RUCO-BAC products can vary extremely, **the customer of RUCO-BAC series himself should examine, test and conduct marketing research** — despite the information, recommendations, test results, sample, care, labelling and processing instructions as well as marketing advice provided by RUDOLF. Therefore, **every customer of RUCO-BAC is responsible for his own specific use, further processing, labelling, marketing...** etc. of substrates finished with RUCO-BAC products. RUDOLF... does not assume any guarantee and/or liability for consequences of examinations and tests or similar omitted by the customer."*

On EPA-acceptable versus non-acceptable claims (verbatim)

*"Every product sold in the United States and advertising with the control and reduction of pests including microorganisms has to be registered by the U.S. Environmental Protection Agency (EPA). Nevertheless, EPA is exempting articles treated with antimicrobial preservatives... from the registration requirement, **if the claims on the labels of treated articles are restricted to the article's preservation itself and do not refer to the consumer's protection against pathogenic organisms.***

"EPA accepts non-public health claims as specified in PR Notice 2000-1.

Examples for acceptable non-public health claims are:

- to inhibit the growth of odour causing bacteria
- to inhibit the growth of mildew causing product deterioration

Examples for non acceptable claims are:

- antibacterial, bactericidal, germicidal
- controls both gram positive and negative bacteria"

On Canada registration status (verbatim)

*"In Canada, **RUCO-BAC AGP, RUCO-BAC AGL, RUCO-BAC HSA CONC, RUCO-BAC CID OF and RUCO-BAC ZPY are not registered by PMRA (Pest Management Regulatory Agency, Canada). Hence, the use of RUCO-BAC products and marketing of textiles treated**" [text continues with restriction language].*

The Master Labels themselves are the smoking gun

EPA Master Labels for RUCO-BAC AGP (84189-2, dated 2016-07-25) and RUCO-BAC AGL (84189-1, dated 2016-07-25) are public PPLS documents. Both labels contain the same explicit caveat on the registered scope:

"Manufactured products incorporating Product EP ... may not make any public health claims."

Public health claims — under EPA's PR Notice 2000-1 framework — include any claim of protection against human pathogens, viruses, or bacteria. The registered scope on Rudolf's own EPA label permits only odor-control and mildew-prevention claims on the finished textile.

Read together with Rudolf's RUCO-BAC disclaimer, the three findings establish five things in writing:

- (1) Rudolf holds the AGP and AGL US EPA registrations (84189-2 and 84189-1, both 2016-07-25).
- (2) Those registrations explicitly prohibit public health claims on manufactured products incorporating the active ingredient.
- (3) Rudolf knows the bright-line test EPA applies — their own RUCO-BAC disclaimer enumerates "antibacterial, bactericidal, germicidal" as NON-acceptable claims for treated articles; only odor-control claims escape registration scrutiny.
- (4) Rudolf admits five of their RUCO-BAC products (AGP, AGL, HSA CONC, CID OF, ZPY) are not registered by Canada's PMRA at all.
- (5) Rudolf legally transfers the responsibility for verifying registration compatibility, conducting marketing research, and ensuring claim-language compliance onto the brand customer — "every customer of RUCO-BAC is responsible for his own specific use, further processing, labelling, marketing."

The competitive lever for FUZE: when a brand evaluates the FUZE alternative, they are evaluating a chemistry that carries federal EPA registration directly under FUZE Biotech's registrant ID, California EPA Q1 2026 approval, OEKO-TEX Standard 100 Class I (infant skin contact), bluesign approval, and PFAS-free verification. There is no disclaimer transferring regulatory risk to the brand customer because there is no regulatory gap to transfer.

Factory Environmental Impact at Point of Manufacture

Every kg of active ingredient that ends up bonded into a finished textile carries the CO₂ footprint of (1) the mining stage where the precious metal was extracted, (2) the refining stage where it was concentrated to industrial grade, (3) the formulation stage where it was mixed into a textile finish, and (4) the global transportation that moved it between each step.

For Rudolf's antimicrobial line, all four stages span multiple continents and grid mixes. For FUZE, all four stages happen on a single floor of a single facility in Salt Lake City.

Rudolf supply chain map (per product)

Product	Mining	Refining	Formulation	Logistics
RUCO-BAC ROX (AGXX silver+ruthenium)	Bushveld Complex, South Africa (deep-shaft, coal grid)	Heraeus Hanau, Germany (~440 g CO2/kWh)	Rudolf GmbH, Geretsried, Germany	ZA → DE → DE → regional mill (triple handoff)
RUCO-BAC AGP / AGL SILVERPLUS (silver chloride)	Bushveld Complex, South Africa (deep-shaft, coal grid)	Aurubis Hamburg or Umicore Olen, Germany / Belgium	Rudolf GmbH, Geretsried, Germany	ZA → DE/BE → DE → regional mill (triple handoff)
Sanitized Puretec (silane-QAC)	Petrochemical feedstock (global naphtha)	Sanitized AG, Burgdorf, Switzerland	Rudolf GmbH, Geretsried, Germany	Global → CH → DE → regional mill (triple handoff)
FUZE F1-F4 (metamaterial)	Recycled electronics feedstock (US-sourced)	FUZE Biotech, Salt Lake City, UT (~300 g CO2/kWh, solar-capable)	FUZE Biotech, Salt Lake City, UT (same facility)	Single-jurisdiction, single facility (no precious metal shipping)

Grid carbon intensity at the formulation stage

Location	Grid carbon intensity (2024)	Source
Geretsried, Germany (Rudolf GmbH)	~440 g CO2 / kWh	Agora Energiewende / Fraunhofer ISE 2024
Hanau, Germany (Heraeus refining)	~440 g CO2 / kWh	Agora Energiewende / Fraunhofer ISE 2024
Burgdorf, Switzerland (Sanitized AG)	~30 g CO2 / kWh	Swiss Federal Office of Energy 2024
Salt Lake City, Utah (FUZE Biotech)	~300 g CO2 / kWh*	US EIA Utah grid mix 2024

*FUZE Biotech production cells are solar-capable; on-site solar generation displaces grid draw during daylight hours, reducing the effective per-kg CO₂ contribution from the formulation stage below the grid average.

The single-site advantage: every kg of FUZE metamaterial reaches a mill having traveled through one facility on one grid in one jurisdiction. Every kg of RUCO-BAC active reaches a mill having traveled through three to four facilities on two to three different grids, plus international shipping. The compounding effect on a brand's Scope 3 emissions inventory is non-trivial — a brand specifying FUZE removes approximately 0.4-0.8 kg CO₂ per kg of fabric treated from their upstream emissions vs. the average Rudolf-line equivalent, before any active-ingredient mining-stage comparison.

Deep Dive: AGXX (RUCO-BAC ROX) and the Ruthenium CO2 Penalty

AGXX is the marketing-headlined innovation in Rudolf’s antimicrobial line. Heraeus and Rudolf entered a strategic partnership in 2023 to bring the silver-ruthenium catalytic technology to textiles. The pitch is compelling: a self-regenerating catalyst that generates reactive oxygen species to kill microorganisms without consuming the precious metal active.

Mechanically the claim is true. The micro-galvanic silver-ruthenium element does work catalytically — the precious metal is not sacrificially consumed during the kill mechanism. But the catalytic-not-consumed framing has been used to imply environmental sustainability that does not survive examination of the upstream supply chain.

Why ruthenium is the highest-CO2 active ingredient in our catalog

Metric	Silver (Ag)	Ruthenium (Ru)	Ratio (Ru:Ag)
Cradle-to-refinery-gate CO2	158 kg CO2/kg	~5,000 kg CO2/kg	31.6×
Recovery yield from ore	85-95% (primary recovery)	<0.1% (PGM byproduct)	~1,000× lower
Primary mining region	Mexico, Peru, China, Poland (diverse)	South African Bushveld (>80% of global supply)	Geographic concentration
Mining method	Open pit + sulfide flotation	Deep-shaft (>1 km depth) plus electrowinning	5-10× energy intensity
Energy source dominance	Mixed (diversified grids)	Coal-heavy South African grid (~880 g CO2/kWh)	2× grid carbon intensity

Sources: Aurubis Environmental Footprint Data 2024 (silver, refinery gate); CRU Group Precious Metals LCA 2024 (ruthenium, Bushveld byproduct); IEA Mining Sector Report 2023 (PGM extraction energy benchmarks).

Per-kg-of-active CO2 math

AGXX at the published 87.5% Ag / 12.5% Ru ratio per the peer-reviewed Frontiers Microbiology and mSphere literature:

Silver contribution: $0.875 \text{ kg Ag} \times 158 \text{ kg CO2/kg} = \mathbf{138 \text{ kg CO2 per kg of AGXX active}}$
Ruthenium contribution: $0.125 \text{ kg Ru} \times 5,000 \text{ kg CO2/kg} = \mathbf{625 \text{ kg CO2 per kg of AGXX active}}$
Total active-ingredient mining CO2: ~763 kg CO2 / kg of AGXX active

Compare a silver-only antimicrobial chemistry (e.g., RUCO-BAC AGP at silver chloride composition): 158 kg CO2 per kg of silver content. AGXX is approximately 4.8× higher per kg of active than the silver-chloride alternative within the same Rudolf product line.

Compare FUZE metamaterial: approximately 38 kg CO2 per kg of active ingredient (FUZE Lifecycle Analysis, 2026). The FUZE metamaterial is produced via liquid laser ablation from recycled electronics feedstock — no traditional mining, no precious-metal refining, no international transportation. The per-kg-active CO2 ratio is approximately **20× lower than AGXX**.

The "self-regenerating catalyst" claim does not reduce this manufacturing CO2 by a single gram. Heraeus still mined every milligram of ruthenium that ends up on the fabric, refined it at Hanau, and shipped it to Rudolf. The catalytic mechanism affects what happens after the textile is in use; it does

not reach back into the supply chain.

Side-by-Side: FUZE vs Each Rudolf Product

Twelve dimensions per comparison. All numeric values are estimated where Rudolf has declined to publish primary data — flagged with the source basis. FUZE values reference internal lifecycle analysis (2026), published third-party test reports (AATCC 100, ASTM E2149, ISO 18184), and federal regulatory documentation.

Dimension	FUZE F1	RUCO-BAC ROX	RUCO-BAC AGP	RUCO-BAC AGL	SILVERPLUS
Active ingredient	FUZE metamaterial	Silver-Ruthenium catalyst (Heraeus AGXX)	Silver chloride	Silver chloride	Silver chloride
Active % w/w (EPA verified)	F1 = 0.0001% on-fabric	Not disclosed (~0.2% est.)	1.75% (EPA 84189-2)	0.20% (EPA 84189-1)	Inherits AGP/AGL
US EPA Reg #	Federal + CA EPA on FUZE Biotech	Not discoverable	84189-2 VERIFIED 2016-07-25	84189-1 VERIFIED 2016-07-25	Inherits AGP/AGL
Registered scope	Full antimicrobial + skin contact	No reg = no scope	Treated-article biocide ONLY; no public health claims permitted	Treated-article biocide ONLY; no public health claims permitted	Same restricted scope as AGP/AGL
Canada PMRA	Compliant	Not registered	Not registered (Rudolf admits)	Not registered (Rudolf admits)	Not registered
Skin contact certified	OEKO-TEX Class I (infant skin)	Not certified for skin contact	Limited; skin-contact apparel marketing exceeds label scope	Marketed for skin-contact apparel (label silent on skin contact)	Marketed for skin-contact apparel
bluesign approved	Yes	No public listing	No public listing	No public listing	No public listing
PFAS-free	Yes	No public statement	No public statement	No public statement	No public statement
Per-kg-active CO2	~38 kg CO2/kg	~763 kg CO2/kg (ruthenium dominant)	~158 kg CO2/kg	~158 kg CO2/kg	~158 kg CO2/kg
Manufacturing location	Salt Lake City, UT (single site)	ZA mining → DE refining → DE formulation (triple jurisdiction)	ZA → DE → DE (triple jurisdiction)	ZA → DE → DE (triple jurisdiction)	ZA → DE → DE (triple jurisdiction)
Grid carbon at formulation site	~300 g CO2/kWh (solar-capable)	~440 g CO2/kWh (German grid)	~440 g CO2/kWh	~440 g CO2/kWh	~440 g CO2/kWh
Wash durability claim	100 washes (F1 Full Spectrum)	Catalyst non-consumed, "Highly resistant" no wash count published (no number)		Skin-contact durable (no number)	50+ washes claimed
Third-party AATCC 100 + E2149 reports public	Yes (on request)	No	No	No	No

Where each Rudolf chemistry has a legitimate role

RUCO-BAC ROX (AGXX): Genuinely interesting catalytic chemistry for industrial applications where the brand can absorb the precious-metal CO2 footprint and where antimicrobial claims are kept narrow (industrial workwear in markets where the product carries the appropriate biocidal registration). Not appropriate for apparel destined for North American sustainability-conscious brand positioning given the ruthenium-driven manufacturing footprint.

RUCO-BAC AGP and AGL: Standard silver-textile chemistry. Comparable to other silver-chloride or silver-polymer textile finishes in the market. The competitive question against FUZE on this product family reduces to (1) the marketing-vs-regulatory-scope gap covered in the EPA audit section, (2) silver wash-leaching ecotoxicity, and (3) the German-grid manufacturing footprint versus a US-manufactured alternative.

SILVERPLUS: Marketed responsibly when scoped to the odor-control claim. The compliance risk surfaces specifically when downstream brand messaging extends into broader antimicrobial framing, at which point the underlying AGP/AGL

registration verification becomes load-bearing.

Sanitized Puretec: Cleanest of the Rudolf-line offerings on US compliance because Sanitized AG carries the registration directly. The distribution-handoff penalty (Switzerland → Germany → regional subsidiary → mill) is the primary sustainability friction; the chemistry itself is a separate evaluation question covered in our dedicated Sanitized comparison.

Sources and Methodology

Every numeric claim in this brief traces to a published source listed below. Where Rudolf has declined to publish primary data, estimates carry the basis citation alongside an explicit "estimated" flag. The audit was conducted on 2026-05-26 using publicly accessible documentation as of that date.

Primary sources — Rudolf Group

- **EPA Master Label — RUCO-BAC AGP (Reg 84189-2, dated 2016-07-25, Silver Chloride 1.75%)**
https://www3.epa.gov/pesticides/chem_search/ppls/084189-00002-20160725.pdf
- **EPA Master Label — RUCO-BAC AGL (Reg 84189-1, dated 2016-07-25, Silver Chloride 0.20%)**
https://www3.epa.gov/pesticides/chem_search/ppls/084189-00001-20160725.pdf
- **Rudolf RUCO-BAC series disclaimer (Version 2022-11-02)**
https://rudolf.com/uploads/rudolfgroup/Documents/disclaimer_ruco_bac_series_en.pdf
- **Rudolf SILVERPLUS technology page**
<https://rudolf.com/technologies/silverplus>
- **RUCO-BAC AGP technical presentation (Rudolf Pakistan)**
https://www.rudolf-group.pk/fileadmin/user_upload/technical_presentation_ruco_bac_agp.pdf
- **Rudolf launches RUCO-BAC AGP coverage (Innovation in Textiles)**
<https://www.innovationintextiles.com/rudolf-launches-durable-antibacterial-and-antiviral-solution-for-textiles/>

Primary sources — Heraeus AGXX technology

- **Heraeus AGXX antimicrobial technology overview**
<https://www.heraeus-precious-metals.com/en/products-solutions/category/antimicrobial-technology/about-agxx/>
- **Heraeus + Rudolf AGXX strategic partnership (Kohan Textile Journal)**
<https://kohantextilejournal.com/heraeus-precious-metals-and-rudolf-form-strategic-partnership-to-bring-agxx-technology-to-the-textile-industry/>
- **AGXX antimicrobial coating mechanism (Frontiers Microbiology, 2018)**
<https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2018.03037/full>
- **Silver-ruthenium antimicrobial mechanism (mSphere, 2025)**
<https://journals.asm.org/doi/10.1128/msphere.00017-25>
- **AGXX aminoglycoside potentiation against P. aeruginosa (mSphere, 2023)**
<https://journals.asm.org/doi/full/10.1128/msphere.00190-23>

Sustainability and LCA references

- Aurubis AG Environmental Footprint Data 2024 (silver refinery-gate CO2)
- CRU Group Precious Metals LCA 2024 (ruthenium byproduct recovery)
- IEA Mining Sector Report 2023 (PGM extraction energy benchmarks)
- Agora Energiewende / Fraunhofer ISE 2024 (German grid carbon intensity)
- Swiss Federal Office of Energy 2024 (Swiss grid mix)
- US Energy Information Administration 2024 (Utah grid mix)
- Reed et al. "Modeling silver release from textiles" Environmental Science & Technology 44(7), 2010 (textile wash leaching baselines)
- Benn & Westerhoff "Nanoparticle silver released into water from commercially available sock fabrics" Environmental Science & Technology 42(11), 2008
- FUZE Biotech Lifecycle Analysis 2026 (internal)
- Phase 19.5 Competitor SDS Audit transcript (deliverables/Competitor_SDS_Audit_2026-05.md)

Methodology notes

Active-ingredient concentrations: Rudolf does not publish % w/w composition for any RUCO-BAC product on their public website. However, direct fetch of the EPA Master Labels for AGP (84189-2, https://www3.epa.gov/pesticides/chem_search/ppls/084189-00002-20160725.pdf) and AGL (84189-1, https://www3.epa.gov/pesticides/chem_search/ppls/084189-00001-20160725.pdf) disclosed the exact registered concentrations: AGP = 1.75% silver chloride, AGL = 0.20% silver chloride. ROX (AGXX) values use peer-reviewed literature for the silver-ruthenium ratio since no Master Label is publicly discoverable. All estimates flagged with the basis citation in our underlying audit transcript.

CO2 footprint calculations: Cradle-to-refinery-gate values use the Aurubis EFD 2024 figure for silver (158 kg CO2/kg refinery-gate) and CRU Group's PGM LCA framework for ruthenium (~5,000 kg CO2/kg refinery-gate, reflecting byproduct allocation from the South African Bushveld Complex). Formulation-stage values combine published grid carbon intensities with the textile chemistry industry's standard energy-per-kg benchmarks. The FUZE per-kg-active CO2 figure derives from FUZE Biotech's 2026 internal lifecycle analysis covering recycled-electronics feedstock, liquid laser ablation production, and the Utah grid mix.

EPA registration verification: Both AGP (84189-2) and AGL (84189-1) registrations were directly verified by fetching the EPA Master Label PDFs from the public PPLS hostname (https://www3.epa.gov/pesticides/chem_search/ppls/) on 2026-05-27. The Master Labels disclose the active concentrations (silver chloride 1.75% and 0.20% respectively) and contain the verbatim restriction quoted in the EPA audit section of this brief. Brands relying on a Rudolf-line product for antimicrobial claims in the US market should pull these Master Label PDFs directly and confirm that their downstream marketing language stays within the registered scope. Marketing language exceeding the registered scope triggers FIFRA Section 12(a)(1)(E) misbranding liability — distinct from Section 12(a)(1)(A) unregistered-pesticide liability which applies to ROX.

Brand voice: All FUZE-side terminology in this brief uses our canonical language (FUZE, metamaterial, F1-F4 tier system). Competitor chemistries are described using the IUPAC names and product names from the competitor's own marketing materials, EPA registrations, and peer-reviewed scientific literature.

Questions about specific claims in this brief? Contact andrew@fuze47.com for the underlying audit transcript, the per-product comparison sheets, the FUZE 2026 Lifecycle Analysis full document, or third-party test reports (AATCC 100, ASTM E2149, ISO 18184) for any FUZE tier.